

Chaitanya Amballa

last updated: September 2026

PhD Student, ECE · University of Illinois Urbana–Champaign (UIUC)

Email: amballa2@illinois.edu

Scholar: [Google Scholar](#)

GitHub: [/achaitu](#)

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Website: achaitanya.web.illinois.edu

Education

University of Illinois Urbana–Champaign

Urbana, IL

Ph.D. in Electrical and Computer Engineering

2022 – present

Advisor: Prof. Romit Roy Choudhury ([SiNRG @ Illinois](#))

Research: generative modeling, spatial inference, diffusion models for inverse problems

University of Illinois Urbana–Champaign

Urbana, IL

M.S. in Electrical and Computer Engineering

2022 – 2025

Advisor: Prof. Romit Roy Choudhury

Indian Institute of Technology Tirupati

Tirupati, India

B.Tech. in Electrical Engineering

2015 – 2019

Advisor: Prof. K. P. Naveen

Honors & Awards

Shun Lien Chuang Memorial Award for Excellence in Graduate Education

2026

Best Poster Award, HotMobile 2026

2026

Qualcomm Graduate Award

2025

Best Bowler Award, Illinois Premier League

2022

Silver Medals, Intra IIT Table Tennis and Cricket

2018

INSPIRE Scholarship

2015

IMMO Silver Medal

2013

Sai Pratibha Puraskar

2013

NSO & NCO Gold Medals

2007 & 2008

Preprints

[1] Generating High-Fidelity Impulse Responses using Conditional Flows

[Chaitanya Amballa](#), Ayush Barik, Haoxiang Lin, Zhen Sun, Mehmet Ergezer, Vamshi Krishna Chillara, Wontak Kim, Romit Roy Choudhury

[2] Can Sparse Walking Patterns Disclose Spatial Maps?

[Chaitanya Amballa](#)^{*}, Sattwik Basu^{*}, Mrudula Athi, Mohamed Mansour, Mehmet Ergezer, Wontak Kim, Romit Roy Choudhury

(^{*} = equal contribution)

[3] Learning Energy-based Variational Latent Prior for VAEs

Debottam Dutta, [Chaitanya Amballa](#), Zhongweiyang Xu, Yu-Lin Wei, Romit Roy Choudhury

Publications

[1] Discrete Langevin-Inspired Posterior Sampling

[Chaitanya Amballa](#)^{*}, Sattwik Basu^{*}, Jorge Vančo Sampedro, Romit Roy Choudhury

(^{*} = equal contribution)

NeurIPS, 2026

- [2] **AVMeme Exam: A Multimodal Multilingual Multicultural Benchmark for LLMs' Contextual and Cultural Knowledge and Thinking** [\[project page\]](#) 
Xilin Jiang, Qiaolin Wang, ..., [Chaitanya Amballa](#)[†], ..., Romit Roy Choudhury, Nima Mesgarani
([†] = co-second author)
COLM, 2026
- [3] **Explicit Context-Driven Neural Acoustic Modeling for High-Fidelity RIR Generation** 
Chen Si, Qianyi Wu, [Chaitanya Amballa](#), Romit Roy Choudhury
Interspeech, 2026
- [4] **Blind Audio Restoration using Contrastive Diffusion Guidance** [\[project page\]](#) 
Sattwik Basu*, [Chaitanya Amballa](#)*, Zhongweiyang Xu, Jorge Vančo Sampedro, Srihari Nelakuditi, Romit Roy Choudhury
(* = equal contribution)
ICML Workshop on ML for Audio, 2026
- [5] **Contrastive Diffusion Guidance for Spatial Inverse Problems** [\[project page\]](#) 
[Chaitanya Amballa](#)*, Sattwik Basu*, Zhongweiyang Xu, Jorge Vančo Sampedro, Srihari Nelakuditi, Romit Roy Choudhury
(* = equal contribution)
ICLR, 2026
- [6] **Inferring Indoor Layouts using Audio** 
Zhijian Yang, Sattwik Basu*, [Chaitanya Amballa](#)*, Debottam Dutta, Srihari Nelakuditi, Romit Roy Choudhury
HotMobile, 2026
- [7] **Can NeRFs See without Cameras?** [\[project page\]](#) 
[Chaitanya Amballa](#), Yu-Lin Wei, Sattwik Basu, Zhijian Yang, Mehmet Ergezer, Romit Roy Choudhury
NeurIPS, 2025
- [8] **Identifying Near-Optimal Decisions in Linear-in-Parameter Bandit Models with Continuous Decision Sets** 
Sanjay P. Bhat, [Chaitanya Amballa](#)
UAI, 2022
- [9] **Computing an Efficient Exploration Basis for Learning with Univariate Polynomial Features** 
[Chaitanya Amballa](#), Manu K. Gupta, Sanjay P. Bhat
AAAI, 2021
- [10] **Coexistence of LTE-Unlicensed and WiFi: A Reinforcement Learning Framework** 
K. P. Naveen, [Chaitanya Amballa](#)
COMSNETS, 2021
- [11] **Learning Algorithms for Dynamic Pricing: A Comparative Study** 
[Chaitanya Amballa](#), Narendhar Gugulothu, Manu K. Gupta, Sanjay P. Bhat
ICML Workshop on Real World Experiment Design and Active Learning, 2020

[12] **Coexistence of LTE-Unlicensed and WiFi: Optimization and Game-Theoretic Frameworks**

Chaitanya Amballa, K. P. Naveen

SPCOM, 2020



Research
Projects

Human Body Pose Estimation with No Fine-Tuning [\[project page\]](#)

Oct 2025 – Dec 2025

Advisor: Dr. Zhizhen Zhao

Diffusion and flow matching for HMD-based pose estimation; posterior sampling for generalization across body shapes

RIR Generation and Geometry Estimation

Feb 2025 – May 2025

Advisor: Dr. Minje Kim

RIR generation using diffusion models; optimizing for unknown source position using Alexa measurements

Click2Hear: Spatial Audio Separation using Visual Cues [\[project page\]](#)

Aug 2022 – Dec 2022

Advisor: Dr. Saurabh Gupta

Unsupervised source separation; binaural audio from video with mono audio

Co-existence of LTE-Unlicensed and WiFi

Aug 2018 – May 2019

Advisor: Dr. K. P. Naveen

Optimization, game-theoretic, and Q-learning frameworks for spectrum coexistence

Speech Enhancement

Aug 2018 – Dec 2018

Advisor: Dr. Rama Krishna Sai Gorthi

Convolutional autoencoder networks for real-time speech denoising

Blood Cell Classification and Disease Diagnosis

Sep 2017 – Jun 2018

Advisors: Dr. Rama Krishna Sai Gorthi & Dr. Subrahmanyam Gorthi

CNN-based classification of blood cells and disease detection from smear images

Industry
Experience

Applied Scientist Intern

May 2026 – Aug 2026

Amazon

Cambridge, MA

Applied Scientist Intern

May 2025 – Aug 2025

Amazon

Cambridge, MA

Researcher

Jul 2019 – Jul 2022

TCS Research

Hyderabad, India

Sequential decision making, stochastic linear bandits, Thompson sampling, regret analysis under

[Dr. Sanjay P. Bhat](#)

Project Associate

Jun 2019 – Jul 2019

IIT Tirupati

Tirupati, India

Research Intern

May 2018 – Jul 2018

Lekha Wireless Solutions

Bengaluru, India

Teaching	Teaching Assistant ECE 498/598: Deep Generative Models <i>Office hours, grading, and student support</i>	University of Illinois Urbana–Champaign Spring 2026
	Teaching Assistant ECE 434: Real-World Algorithms for IoT and Data Science <i>Office hours, grading, and student support</i>	University of Illinois Urbana–Champaign Spring 2024, Spring 2025
Skills	Generative Modeling: Diffusion Models, NeRFs, Flow Matching, VAEs Frameworks: PyTorch, TensorFlow, HuggingFace Diffusers/Transformers Tooling: Slurm, Weights & Biases, Git, Streamlit, $\LaTeX$ Programming: Python, C/C++, MATLAB, SQL	
Selected Coursework	ECE 598: Deep Generative Models ECE 598: Generative Modeling with Diffusion and Flow Matching Models CS 598: Generative Models for Audio ECE 549: Computer Vision ECE 543: Statistical Learning Theory ECE 586: MDPs and Reinforcement Learning ECE 534: Random Processes ECE 490: Introduction to Optimization	
Thesis	[1] M.S. Thesis Chaitanya Amballa <i>M.S. in Electrical and Computer Engineering, UIUC, 2025</i>	
	[2] B.Tech. Thesis Chaitanya Amballa <i>B.Tech. in Electrical Engineering, IIT Tirupati, 2019</i>	
Leadership	House Captain, Team “Sivaliks” Member, Technical Committee Electrical Engineering Department Sports Representative	IIT Tirupati, 2016–2017 IIT Tirupati, 2017–2019 IIT Tirupati, 2017–2018